```
======================================================================
TIER 3 ANALYSIS: Era & Vaccination Stratification
  Model per stratum: logit(IHD) = Age + Male + CCI
======================================================================
Loaded cohort: 556,764 patients from cohort_tier3_ready.csv
Groups: {'Group 2': 483981, 'Group 3': 70838, 'Group 1': 1870, 'Unknown': 75}
Preparing analysis data...
  CCI computed from 13/14 components (excluding ['Myocardial_Infarction', 'Congestive_Heart_Failure', 'AIDS_HIV'])
  CCI validation (G1/G2): 99.0% match with Step 8 scores
  G3 CCI sanity check: mean=1.34 (should be ~3.5 per Tier 2 results)
  Race distribution: {'Chinese': 252173, 'Indian': 111777, 'Others': 71628, 'Malay': 70847}
COVID cohort: 467,352 (G1=1,826, G2=465,526)
  Ancestral: N=60,619, G1=58
  Delta: N=207,893, G1=417
  Omicron: N=198,840, G1=1,351

======================================================================
COMPONENT A: Era-Stratified Logistic Regression (G1 vs G2)
  Model: logit(IHD) = Age + Male + CCI, separately per era
======================================================================

  --- Ancestral Era ---
  N=60,619 (G1=58, G2=60,561, rate=0.10%)
  Dropped 627 rows with missing race
  [G1G2_Ancestral] Including race (4 levels, ref=Chinese, coverage=100%)
  [G1G2_Ancestral] Events=57 < 80: using Firth's penalized regression
  [G1G2_Ancestral] Firth approximation via penalized ML (small-sample correction)
  Fit method: firth_approx
  Apparent AUC: 0.8478 (training-set, optimistically biased)
  Pseudo R2: 0.113923
  AIC: 817.90
    Intercept              OR=0.0000  (0.0000-0.0000)  p=9.23e-38 ***
    C(race, Treatment(reference="Chinese"))[T.Indian]  OR=3.5982  (1.0299-12.5712)  p=4.48e-02 *
    C(race, Treatment(reference="Chinese"))[T.Malay]  OR=7.3183  (1.6681-32.1074)  p=8.34e-03 **
    C(race, Treatment(reference="Chinese"))[T.Others]  OR=5.1623  (1.5384-17.3224)  p=7.88e-03 **
    age                    OR=1.1103  (1.0877-1.1334)  p=2.16e-23 ***
    gender_male            OR=1.6279  (0.5775-4.5885)  p=3.57e-01
    cci_score              OR=1.0411  (0.6955-1.5585)  p=8.45e-01

  --- Delta Era ---
  N=207,893 (G1=417, G2=207,476, rate=0.20%)
  Dropped 4 rows with missing race
  [G1G2_Delta] Including race (4 levels, ref=Chinese, coverage=100%)
  Fit method: bfgs
  Apparent AUC: 0.8825 (training-set, optimistically biased)
  Pseudo R2: 0.152503
  AIC: 5110.59
    Intercept              OR=0.0000  (0.0000-0.0000)  p=0.00e+00 ***
    C(race, Treatment(reference="Chinese"))[T.Indian]  OR=1.9631  (1.5088-2.5542)  p=5.09e-07 ***
    C(race, Treatment(reference="Chinese"))[T.Malay]  OR=1.8040  (1.3884-2.3439)  p=1.00e-05 ***
    C(race, Treatment(reference="Chinese"))[T.Others]  OR=0.2543  (0.0875-0.7389)  p=1.19e-02 *
    age                    OR=1.0738  (1.0664-1.0813)  p=5.72e-90 ***
    gender_male            OR=2.5139  (1.9977-3.1637)  p=3.84e-15 ***
    cci_score              OR=1.2644  (1.1951-1.3378)  p=3.40e-16 ***

  --- Omicron Era ---
  N=198,840 (G1=1,351, G2=197,489, rate=0.68%)
  Dropped 43 rows with missing race
  [G1G2_Omicron] Including race (4 levels, ref=Chinese, coverage=100%)
  Fit method: bfgs
  Apparent AUC: 0.8457 (training-set, optimistically biased)
  Pseudo R2: 0.141240
  AIC: 13908.45
    Intercept              OR=0.0001  (0.0001-0.0002)  p=0.00e+00 ***
    C(race, Treatment(reference="Chinese"))[T.Indian]  OR=1.6516  (1.3791-1.9779)  p=4.91e-08 ***
    C(race, Treatment(reference="Chinese"))[T.Malay]  OR=1.8808  (1.6290-2.1716)  p=7.15e-18 ***
    C(race, Treatment(reference="Chinese"))[T.Others]  OR=0.4945  (0.3183-0.7681)  p=1.73e-03 **
    age                    OR=1.0532  (1.0494-1.0570)  p=3.97e-174 ***
    gender_male            OR=1.8260  (1.6329-2.0420)  p=4.61e-26 ***
    cci_score              OR=1.2180  (1.1860-1.2508)  p=8.15e-48 ***

  Era model summary:
     Era      N  Events  Rate_pct   EPV  N_predictors  Race_included  Firth_used  Apparent_AUC   Age_OR  Male_OR   CCI_OR Status
ncestral  60619      58  0.095680   9.7             6           True        True      0.847823 1.110325 1.627867 1.041104     OK
   Delta 207893     417  0.200584  69.5             6           True       False      0.882510 1.073789 2.513946 1.264434     OK
 Omicron 198840    1351  0.679441 225.2             6           True       False      0.845700 1.053168 1.826047 1.217997     OK
```

```
======================================================================
COMPONENT B: Vaccination-Stratified Models (within Delta & Omicron)
  Ancestral era excluded (vaccines not yet available)
  Vaccination window: any dose within 6 months before COVID date
======================================================================
  Using 6-month vaccination window (vaccinated_6mo_before_covid)

 Delta: Vaccinated=161,680, Unvaccinated=46,213 (using vaccinated_6mo_before_covid)

 --- Delta / Unvaccinated ---
  N=46,213 (G1=87, rate=0.19%)
  [G1G2_Delta_Unvaccinated] Including race (4 levels, ref=Chinese, coverage=100%)
  Apparent AUC: 0.9201 (training-set), Pseudo R2: 0.199874
    Intercept             OR=0.0000  (0.0000-0.0000) p=6.20e-85 ***
    C(race, Treatment(reference="Chinese"))[T.Indian]  OR=1.0881  (0.5571-2.1253) p=8.05e-01
    C(race, Treatment(reference="Chinese"))[T.Malay]  OR=1.7049  (0.9662-3.0082) p=6.56e-02
    C(race, Treatment(reference="Chinese"))[T.Others]  OR=0.0000  (0.0000-186907128101756940092734205954401684012644176389681761406516751536396486705026170 88.0000) p=9.19e-01
    age                   OR=1.0780  (1.0628-1.0934) p=4.28e-25 ***
    gender_male           OR=2.7676  (1.6662-4.5970) p=8.43e-05 ***
    cci_score             OR=1.1442  (1.0085-1.2981) p=3.64e-02 *

 --- Delta / Vaccinated ---
  N=161,680 (G1=330, rate=0.20%)
  [G1G2_Delta_Vaccinated] Including race (4 levels, ref=Chinese, coverage=100%)
  Apparent AUC: 0.8722 (training-set), Pseudo R2: 0.142110
    Intercept             OR=0.0000  (0.0000-0.0000) p=7.56e-310 ***
    C(race, Treatment(reference="Chinese"))[T.Indian]  OR=2.2190  (1.6581-2.9698) p=8.29e-08 ***
    C(race, Treatment(reference="Chinese"))[T.Malay]  OR=1.8889  (1.4082-2.5337) p=2.19e-05 ***
    C(race, Treatment(reference="Chinese"))[T.Others]  OR=0.4837  (0.1972-1.1867) p=1.13e-01
    age                   OR=1.0727  (1.0642-1.0813) p=2.77e-66 ***
    gender_male           OR=2.4608  (1.9002-3.1868) p=8.69e-12 ***
    cci_score             OR=1.3002  (1.2210-1.3847) p=2.85e-16 ***

 Omicron: Vaccinated=129,532, Unvaccinated=69,308 (using vaccinated_6mo_before_covid)

 --- Omicron / Unvaccinated ---
  N=69,308 (G1=358, rate=0.52%)
  [G1G2_Omicron_Unvaccinated] Including race (4 levels, ref=Chinese, coverage=100%)
  Apparent AUC: 0.9025 (training-set), Pseudo R2: 0.203814
    Intercept             OR=0.0001  (0.0001-0.0002) p=2.80e-271 ***
    C(race, Treatment(reference="Chinese"))[T.Indian]  OR=2.0509  (1.5120-2.7819) p=3.87e-06 ***
    C(race, Treatment(reference="Chinese"))[T.Malay]  OR=1.6807  (1.2542-2.2522) p=5.07e-04 ***
    C(race, Treatment(reference="Chinese"))[T.Others]  OR=0.1913  (0.0702-0.5209) p=1.21e-03 **
    age                   OR=1.0550  (1.0486-1.0615) p=1.24e-66 ***
    gender_male           OR=2.0000  (1.6080-2.4876) p=4.74e-10 ***
    cci_score             OR=1.1991  (1.1394-1.2620) p=3.32e-12 ***

 --- Omicron / Vaccinated ---
  N=129,532 (G1=993, rate=0.77%)
  [G1G2_Omicron_Vaccinated] Including race (4 levels, ref=Chinese, coverage=100%)
  Apparent AUC: 0.8161 (training-set), Pseudo R2: 0.115873
    Intercept             OR=0.0001  (0.0001-0.0002) p=0.00e+00 ***
    C(race, Treatment(reference="Chinese"))[T.Indian]  OR=1.4623  (1.1677-1.8311) p=9.29e-04 ***
    C(race, Treatment(reference="Chinese"))[T.Malay]  OR=1.9477  (1.6511-2.2977) p=2.63e-15 ***
    C(race, Treatment(reference="Chinese"))[T.Others]  OR=0.7154  (0.4375-1.1696) p=1.82e-01
    age                   OR=1.0524  (1.0477-1.0571) p=8.27e-112 ***
    gender_male           OR=1.7845  (1.5666-2.0327) p=2.88e-18 ***
    cci_score             OR=1.2221  (1.1846-1.2608) p=1.89e-36 ***

 Vaccination model summary:
   Era  Vaccination      N  Events  Rate_pct      AUC   CCI_OR Status
 Delta Unvaccinated  46213      87  0.188259 0.920140 1.144191     OK
 Delta   Vaccinated 161680     330  0.204107 0.872248 1.300238     OK
Omicron Unvaccinated  69308     358  0.516535 0.902458 1.199101     OK
Omicron   Vaccinated 129532     993  0.766606 0.816070 1.222126     OK
```

```
======================================================================
COMPONENT C: Era-Stratified G1 vs G3 (COVID Exposure Model)
  logit(COVID_IHD) = Age + Male + CCI, per era
  G3 (all eras) serves as the common reference group
  NOTE: Vaccination excluded from G1 vs G3 - reference dates differ
        conceptually (G1: pre-COVID, G3: pre-IHD). See biostats review Q9.
======================================================================
  G3 reference pool: 6,439 patients

 --- Ancestral: G1=58 vs G3=6,439 ---
  [G1G3_Ancestral] Events=58 < 80: using Firth's penalized regression
  [G1G3_Ancestral] Firth approximation via penalized ML (small-sample correction)
  Apparent AUC: 0.9052 (training-set, optimistically biased)
    Intercept            OR=4.2188  (0.8941-19.9064)  p=6.90e-02
    age                  OR=0.9120  (0.8903-0.9343)  p=6.96e-14 ***
    gender_male          OR=1.1318  (0.4372-2.9302)  p=7.99e-01
    cci_score            OR=0.3154  (0.1861-0.5344)  p=1.80e-05 ***
  INTERPRETATION: CCI OR=0.315 (INVERSE). Ancestral-era COVID-IHD patients are HEALTHIER than non-COVID IHD - supports COVID-specific pathway.

 --- Delta: G1=417 vs G3=6,439 ---
  Apparent AUC: 0.6172 (training-set, optimistically biased)
    Intercept            OR=0.2521  (0.1413-0.4499)  p=3.12e-06 ***
    age                  OR=0.9844  (0.9768-0.9920)  p=6.30e-05 ***
    gender_male          OR=0.9939  (0.7778-1.2699)  p=9.61e-01
    cci_score            OR=0.8536  (0.8059-0.9042)  p=7.11e-08 ***
  INTERPRETATION: CCI OR=0.854 (INVERSE). Delta-era COVID-IHD patients are HEALTHIER than non-COVID IHD - supports COVID-specific pathway.

 --- Omicron: G1=1,351 vs G3=6,439 ---
  Apparent AUC: 0.6274 (training-set, optimistically biased)
    Intercept            OR=0.0281  (0.0189-0.0417)  p=1.38e-70 ***
    age                  OR=1.0312  (1.0262-1.0362)  p=3.60e-35 ***
    gender_male          OR=0.7928  (0.6966-0.9023)  p=4.33e-04 ***
    cci_score            OR=0.9848  (0.9569-1.0135)  p=2.96e-01
  INTERPRETATION: CCI NOT significant - similar comorbidity between COVID-IHD and non-COVID IHD.

 --- SENSITIVITY: Calendar-Matched G3 ---
  Restricting G3 to IHD events within each era's calendar window

 Ancestral: G1=58 vs G3(matched)=5,463
  [G1G3_matched_Ancestral] Events=58 < 80: using Firth's penalized regression
  [G1G3_matched_Ancestral] Firth approximation via penalized ML (small-sample correction)
  CCI OR=0.3120 (0.1839-0.5295) p=1.59e-05
  vs pooled G3: OR delta = 0.0033 (<10% shift)

 Delta: G1=417 vs G3(matched)=678
  CCI OR=0.8711 (0.8154-0.9306) p=4.21e-05
  vs pooled G3: OR delta = 0.0174 (<10% shift)

 Omicron: G1=1,351 vs G3(matched)=298
  CCI OR=0.9812 (0.9241-1.0418) p=5.35e-01
  vs pooled G3: OR delta = 0.0036 (<10% shift)
  Saved: g1g3_pooled_vs_matched_comparison.csv
```

```
======================================================================
COMPONENT D: Severity as Exploratory Covariate (CAUSAL CAVEAT)
WARNING: Severity may be a mediator. Results are exploratory.
======================================================================
Patients with known severity: 458,835 (G1=1,797)
Severity distribution:
severity_category
Moderate    256109
Mild        195908
Severe        4905
Critical      1913

Severity model (dummy variables, ref=Mild):
  Intercept                                        OR=0.0001  (0.0001-0.0001)  p=0.00e+00 ***
  C(severity_category, Treatment(reference="Mild"))[T.Critical]  OR=2.2629  (1.7649-2.9014)  p=1.20e-10 ***
  C(severity_category, Treatment(reference="Mild"))[T.Moderate]  OR=0.9845  (0.8807-1.1005)  p=7.83e-01
  C(severity_category, Treatment(reference="Mild"))[T.Severe]  OR=2.4425  (2.0673-2.8859)  p=9.21e-26 ***
  age                                              OR=1.0593  (1.0560-1.0625)  p=2.89e-295 ***
  gender_male                                      OR=1.8451  (1.6689-2.0398)  p=5.31e-33 ***
  cci_score                                        OR=1.2399  (1.2105-1.2701)  p=8.98e-69 ***

Binary ICU model: 867 ICU admissions in severity subset
Binary ICU model results:
  Intercept                  OR=0.0001  (0.0001-0.0001)  p=0.00e+00 ***
  age                        OR=1.0617  (1.0586-1.0648)  p=0.00e+00 ***
  gender_male                OR=1.8526  (1.6760-2.0478)  p=1.65e-33 ***
  cci_score                  OR=1.2478  (1.2183-1.2780)  p=1.68e-73 ***
  icu_admission              OR=5.0058  (3.7180-6.7399)  p=2.59e-26 ***
[sev_Ancestral] Events=47 < 80: using Firth's penalized regression
[sev_Ancestral] Firth approximation via penalized ML (small-sample correction)

--- LOS as Continuous Covariate ---
Patients with LOS data: 466,678 (G1=1,825)
LOS distribution: mean=8.5, median=8, IQR=6-9
LOS continuous model results:
  Intercept                  OR=0.0001  (0.0000-0.0001)  p=0.00e+00 ***
  age                        OR=1.0619  (1.0588-1.0650)  p=0.00e+00 ***
  gender_male                OR=1.8570  (1.6810-2.0515)  p=3.80e-34 ***
  cci_score                  OR=1.2566  (1.2271-1.2867)  p=1.61e-79 ***
  LOS                        OR=1.0159  (1.0101-1.0217)  p=6.01e-08 ***

LOS by Group x Era:
Group       Era      N  LOS_mean  LOS_median  LOS_q25  LOS_q75  LOS_max
Group 1 Ancestral     57 14.561404        16.0      3.0     21.0     40.0
Group 1     Delta    417 11.335731         9.0      8.0     11.0    115.0
Group 1   Omicron   1351  7.133235         6.0      5.0      8.0     35.0
Group 2 Ancestral  59935 15.174322        18.0     10.0     20.0    334.0
Group 2     Delta 207472  8.803164         8.0      8.0      9.0    165.0
Group 2   Omicron 197446  6.181047         6.0      5.0      6.0     92.0
```

```
--- Severity Distribution by Group x Era ---

Severity distribution:
group variant_era  Critical     Mild  Moderate  Severe  Unknown
Group 1   Ancestral       0.0      7.0      40.0     0.0     11.0
Group 1       Delta      21.0     60.0     278.0    48.0     10.0
Group 1     Omicron      51.0    909.0     256.0   127.0      8.0
Group 2   Ancestral      69.0   8109.0   47614.0     5.0   4764.0
Group 2       Delta     889.0  19197.0  182582.0  2060.0   2748.0
Group 2     Omicron     883.0 167626.0   25339.0  2665.0    976.0

Event rate by severity:
Severity      N  G1_events  Event_rate_pct
    Mild 195908        976        0.498193
Moderate 256109        574        0.224123
  Severe   4905        175        3.567788
Critical   1913         72        3.763722
 Unknown   8517         29        0.340495
```

```
======================================================================
COMPONENT E: Interaction Tests (Era  x  CCI)
======================================================================
  Pooled sample: 467,352 (events=1,826)
  Reference category: Omicron (N_G1=198,840)

 LR Test (interaction vs no interaction):
    Chi-square: 12.8153
    df: 2
    p-value: 1.6489e-03
    SIGNIFICANT: Era modifies the CCI-IHD association.

 Full interaction model:
    Intercept                          OR=0.0001  (0.0001-0.0001)  p=0.00e+00 ***
    age                                OR=1.0559  (1.0526-1.0592)  p=3.97e-258 ***
    gender_male                        OR=1.9381  (1.7544-2.1412)  p=9.37e-39 ***
    cci_score                          OR=1.2186  (1.1868-1.2512)  p=1.27e-48 ***
    era_Ancestral                      OR=0.5790  (0.4318-0.7765)  p=2.63e-04 ***
    era_Delta                          OR=0.5544  (0.4752-0.6469)  p=6.69e-14 ***
    cci_score:era_Ancestral            OR=1.2027  (0.8769-1.6496)  p=2.52e-01
    cci_score:era_Delta                OR=1.1041  (1.0445-1.1670)  p=4.69e-04 ***

 --- Vaccination  x  CCI Interaction Test ---
    Vacc x CCI LR test: chi2=1.6794, p=1.9500e-01

======================================================================
SENSITIVITY: Race as Additional Covariate (G1 vs G2)
======================================================================
  Patients with race data: 466,678 (G1=1,825)
  Race distribution:
race
Chinese    232359
Indian     106015
Malay       65136
Others      63168
  CCI OR without race: 1.2567
  CCI OR with race:    1.2451
  Attenuation: 0.9%
  CONCLUSION: <10% attenuation - race is not a meaningful confounder of CCI-IHD

 Race model ORs:
    Intercept                                    OR=0.0000  (0.0000-0.0001)  p=0.00e+00 ***
    C(race, Treatment(reference="Chinese"))[T.Indian]  OR=1.5764  (1.3615-1.8252)  p=1.15e-09 ***
    C(race, Treatment(reference="Chinese"))[T.Malay]  OR=1.9480  (1.7191-2.2075)  p=1.42e-25 ***
    C(race, Treatment(reference="Chinese"))[T.Others]  OR=0.7613  (0.5726-1.0121)  p=6.05e-02
    age                                          OR=1.0648  (1.0615-1.0681)  p=0.00e+00 ***
    gender_male                                  OR=1.8896  (1.7105-2.0874)  p=5.26e-36 ***
    cci_score                                    OR=1.2451  (1.2158-1.2750)  p=6.91e-73 ***

======================================================================
DESCRIPTIVE TABLES
======================================================================
  Saved: table1_by_era.csv

   Group      N  Age_mean   Age_std  Male_pct  CCI_mean  CCI_median  Vaccinated_pct  Mean_doses  Sev_Critical_pct  Sev_Severe_pct  Sev_Moderate_pct  Sev_Mild_pct  Sev_Unknown_pct
Ancestral  60619 34.762616  9.076400 95.590491  0.020522         0.0        0.090731    0.001485          0.113826        0.008248         78.612316     13.388542         7.877068
    Delta 207893 40.703328 18.307057 63.346529  0.303839         0.0       88.172281    1.828609          0.437725        1.013983         87.958710      9.262938         1.326644
  Omicron 198840 44.786170 27.601308 49.344196  0.776428         0.0       80.242406    2.219709          0.469724        1.404144         12.872159     84.759103         0.494870
  Saved: table1_by_era_group.csv
  Saved: vacc_coverage_by_era.csv

variant_era      N  N_vacc_anytime  Mean_doses  N_vacc_6mo  Pct_vacc_anytime  Pct_vacc_6mo
  Ancestral  60619              55    0.001485          55               0.1           0.1
      Delta 207893          183304    1.828609      161680              88.2          77.8
    Omicron 198840          159554    2.219709      129532              80.2          65.1
```

```
======================================================================
TIER 3: EXECUTIVE SUMMARY
======================================================================
  Saved: ./data/03_results/step_11_tier3/tier3_summary_report.txt
TIER 3 ANALYSIS: EXECUTIVE SUMMARY
======================================================================

MODEL: logit(IHD) = Age + Male + CCI, stratified by variant era & vaccination
TOTAL COVID COHORT: 467,352 (G1=1,826, G2=465,526)
OVERALL EVENT RATE: 0.391%

ANALYSIS CLASSIFICATION
--------------------------------------------------
  CONFIRMATORY (primary inferential tests):
    - Component A: Era-stratified G1 vs G2 models
    - Component E: Era x CCI interaction LR test (2 df)
  EXPLORATORY (descriptive / hypothesis-generating):
    - Component B: Vaccination-stratified models
    - Component C: G1 vs G3 COVID exposure model
    - Component D: Severity as covariate (potential mediator)

EVENT RATES PER STRATUM
--------------------------------------------------
  Ancestral: N=60,619, G1=58, rate=0.10%, EPV=19.3 [Firth penalized]
  Delta: N=207,893, G1=417, rate=0.20%, EPV=139.0
  Omicron: N=198,840, G1=1,351, rate=0.68%, EPV=450.3

A. ERA-STRATIFIED RESULTS (G1 vs G2) [CONFIRMATORY]
--------------------------------------------------
    Era      N  Events  Rate_pct   EPV  N_predictors  Race_included  Firth_used  Apparent_AUC   Age_OR  Male_OR   CCI_OR Status
Ancestral  60619      58  0.095680   9.7             6           True        True      0.847823 1.110325 1.627867 1.041104     OK
   Delta 207893     417  0.200584  69.5             6           True       False      0.882510 1.073789 2.513946 1.264434     OK
  Omicron 198840    1351  0.679441 225.2             6           True       False      0.845700 1.053168 1.826047 1.217997     OK

CCI OR TREND ACROSS ERAS:
  Ancestral: CCI OR = 1.0411 (p = 8.45e-01)
  Delta: CCI OR = 1.2644 (p = 3.40e-16)
  Omicron: CCI OR = 1.2180 (p = 8.15e-48)
  TREND: CCI OR INCREASING across eras
  -> Later variants disproportionately affect sicker patients

C. G1 vs G3 (COVID EXPOSURE MODEL) BY ERA [EXPLORATORY]
--------------------------------------------------
  NOTE: Vaccination excluded from G1 vs G3 (different reference date semantics)
  Ancestral: CCI OR = 0.3154 (0.1861-0.5344) p=1.80e-05
  Delta: CCI OR = 0.8536 (0.8059-0.9042) p=7.11e-08
  Omicron: CCI OR = 0.9848 (0.9569-1.0135) p=2.96e-01
  (Tier 2 overall: CCI OR = 0.704)

KNOWN LIMITATIONS
--------------------------------------------------
  - AUC values are apparent (training-set), optimistically biased
  - Ancestral era has low EPV - Firth penalization applied but interpret with caution
  - G3 pooled across all calendar periods (sensitivity: calendar-matched G3 also run)
  - Immortal time bias possible in G2 (365-day survival requirement)
  - No multiplicity correction applied (pre-specified stratification, not data-dredging)
```